

Drug abuse: why people start using drugs and what kind of effect do drugs have on our brain?

Drug Abuse: More Than Just a Bad Habit — A Deep Dive Into a Global Health Crisis

Drug abuse is commonly mistakenly thought to be merely a lack of self-control or a bad life choice. Actually, it is a complex disorder that affects millions of people across the globe, irrespective of social, economic, and cultural backgrounds. Drug overdose deaths alone surpassed 106,000 in 2021 in the United States (Centers for Disease Control and Prevention, 2022). But what, actually, is drug abuse? This writing specifically defines drug abuse as not merely drug misuse but as a multifaceted psychological, social, and neurological disease that extensively alters the human brain and behavior. To understand it better, we must learn about its causes, its impact, and how it is different from other diseases.

Consider the story of Jason, a high school student with promising potential and an athlete who began using prescription painkillers after suffering an injury while playing sports. What started as legitimate treatment became an addiction that he could not end. Within a year, Jason lost his scholarship, quit school, and was in rehab. Throughout America, there are thousands of young, talented, and promising individuals who become addicted to drugs through a process of exposure, vulnerability, and poor guidance. Jason's case is merely a mirror to a larger reality: drug addiction never begins in malice, but rather during a moment of hurt, stress, or even medical consultation with no supervision.

The Anatomy of Drug Abuse

At its core, drug abuse refers to the compulsive use of substances—legal or illegal—in a way that is harmful to the user or those around them. But this surface-level definition barely scratches the surface. Drug abuse consists of a constellation of interrelated factors: psychological vulnerability, social exposure, environment, and heredity. Psychologically, people also turn to drugs of abuse as a means of self-treatment for underlying untreated mental illness like

depression, anxiety, or post-traumatic stress disorder (PTSD). Socially, peer pressure, particularly in teenagers, has a critical role to play.

Classification: More Than a Bad Habit — A Disease of the Brain

Drug abuse is a type of chronic brain disease. Like diabetes or hypertension, it requires ongoing care and treatment. It's medically classified as a "substance use disorder," underscoring its position not as a moral failing, but as a diagnosable, treatable disease. What sets drug abuse apart from other risky behaviors is its takeover of the brain's reward system. Drugs such as opioids and methamphetamine overwhelm the brain with dopamine, conditioning it to desire the drug more than anything else—more than food, sleep, or relationships. Drug abuse, unlike casual or recreational use, entails ongoing, compulsive pursuit of the drug despite awareness of the damage it inflicts. This is different from, for example, social drinking or prescribed medication use.

Why People Begin Using Drugs

The causes of drug addiction are complex, with a mix of psychological, social, genetic, and environmental factors. Peer pressure and social environment are two of the most prevalent causative factors, especially in teenagers. Teenagers become drug dependent out of curiosity or because of their need to fit into peer groups. Approximately 1 in 8 youths aged 12 to 17 used illicit drugs in 2021 (Substance Abuse and Mental Health Services Administration, 2022). Mental illness is another significant cause of drug misuse. People who suffer from depression, anxiety, trauma, or other mental illnesses can use drugs as a way of self-medication. Approximately 50% of individuals who suffer from a substance use disorder in their lifetime will also suffer from a co-occurring mental illness, like PTSD or bipolar disorder (National Institute on Drug Abuse, 2020).

Although drugs will frequently relieve symptoms short-term, in the long term they typically intensify the existing psychological suffering and create a toxic cycle of dependence. Genetic predisposition and family dynamics are also important. Children who grow up in environments where drug use is the norm or where supervision is poor are at higher risk. In addition, 40% to 60% of an individual's susceptibility to addiction is due to genetics (National Institute on Drug

Abuse, 2017). For instance, genetic difference in the DRD2 gene for the dopamine receptor has been associated with increased risk for substance use disorder, illustrating how neurobiology can predispose individuals to more quickly develop addiction. Drug use initiation in adolescence, when brain development is ongoing, is especially troubling.

Neurologically, the brain of an adolescent is more vulnerable to the rewarding properties of drugs and less capable of taking into account long-term consequences. This unbalanced development of the brain increases the risk for habitual use and long-term addiction.

What Drug Abuse Is Not: Dispelling Misconceptions

It's necessary to define drug abuse not only by what it is—but by what it is not. It is not a sign of weakness. It is not a choice people make lightly. And it is certainly not something that can be "cured" by willpower alone. Drawing these distinctions serves to break the stigma of addiction and substitute judgment and punishment with empathy and support.

The Brain Under Siege: Neurological Effect of Drug Abuse

Drug abuse profoundly changes brain function and structure. Most drugs target the reward system of the brain by artificially raising the levels of dopamine—a neurotransmitter that is important in pleasure, motivation, and learning. Overstimulation of the reward pathway reinforces behavior that results in drug use repetition.

Over time, regular drug use causes the brain to adapt by reducing the level of naturally produced dopamine or by reducing the number of dopamine receptors. This creates a state known as tolerance, in which the users require larger doses to experience the same effect. Also, because natural dopamine production is suppressed, individuals find it more difficult to experience pleasure in normal activities like social interaction or eating, further reinforcing drug-seeking behavior.

Brain imaging research has given us concrete proof of the drug-induced changes in the brain. Dr. Nora Volkow's study demonstrated that cocaine addicts have significantly lower availability of dopamine receptors in the striatum—a key region for motivation and reinforcement (Volkow et

al. 1997). PET and functional MRI scans show decreased activity of the prefrontal cortex, the part of the brain which regulates judgment, self-control, and decision-making.

Chronic drug exposure also impacts the hippocampus, a region involved in memory and learning, and the amygdala, a region involved in the control of emotions and stress response. It is because of these changes that chronic drug users typically have problems with memory, mood swings, impulse control, and stress sensitivity.

In addition, stimulants such as methamphetamine and cocaine have been shown to produce long-lasting adaptations in synaptic connectivity, most notably in the prefrontal cortex and nucleus accumbens. The changes in synaptic plasticity account for the perpetuation of addictive behavior despite episodes of abstinence.

Comparison: How Drug Abuse is Different from Other Disorders

Unlike mood disorders like depression, which can have no external stimuli, drug usage tends to start with exposure—social, environmental, or pharmaceutical. It also differs from habits like overeating or gambling in that chemicals taken can create instant chemical changes in brain chemistry with even single-use events. Within

Real-World Impact: Broader Social and Economic Implications

Beyond the neurological and psychological toll, drug addiction carries immense social and economic implications. Addiction devastates families, leading to relationship dissolution, financial destruction, and emotional pain. The Kaiser Family Foundation discovered in 2023 that nearly 1 in 10 American adults had lost a family member due to drug overdose, and almost a third had a close family member who had struggled with substance addiction.

Economically, drug abuse drains the United States of more than \$740 billion each year in productivity loss, healthcare, and criminal justice, as per NIDA. Communities, especially rural and underserved communities, disproportionately bear the brunt of drug abuse due to their lack of access to prevention and treatment services. Rural America, for example, has been particularly

devastated by the opioid epidemic, with 42% of rural Americans reporting family or personal experience with opioid addiction, as opposed to 30% in suburban areas.

Conclusion

Addiction to drugs is a complex issue with roots in psychological vulnerability, social influence, genetic predisposition, and environmental exposure. Its impact on the brain is deep and long-lasting, interfering with the neurological processes that regulate reward, impulse management, learning, and emotional stability. As the scientific and public health communities persist in understanding the intricacies of addiction, society must invest in large-scale prevention, early intervention, and easily accessible treatment initiatives. Studying the science of addiction is not an ivory tower pursuit—it is a prerequisite to stemming the escalating wave of drug and alcohol abuse and bringing hope to the millions of individuals and families who suffer.

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